

LINEAR LEAST-SQUARES UNMIXING ON A SAMPLE NETWORK: DERIVATION, REGULARIZATION, AND ERROR PROPAGATION

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ABSTRACT

Barnes and Lipp (2024) recover source concentrations on a river sample network by convex optimization, penalising *relative* differences between predicted and observed concentrations. Their Appendix A observes that penalising *absolute* differences instead reduces the problem to a linear least-squares matrix inversion. This note works that observation out in full. We show that the mixing matrix $\mathbf{M}$ is square, row-stochastic and triangular, hence always invertible, and we give its inverse in closed form as a local differencing stencil on the network. We then extend the formulation in three directions that the appendix does not cover: first-order tracer decay, weighting by the data covariance (generalised least squares), and Tikhonov regularization of the model variance. We prove that the regularized problem has a unique solution for every regularization strength, derive the resulting linear estimator, and obtain closed-form covariances for both the recovered sources and the modelled observations. We distinguish carefully between the propagated covariance and the Bayesian posterior covariance, which coincide only when the problem is unregularized, and give the exact identity relating them. Everything is derived from first principles so that it can be checked line by line.

1 SCOPE AND NOTATION

We adopt the notation of Barnes and Lipp (2024) throughout. The sample network is a directed acyclic graph $G = (N, E)$ whose n nodes are sample sites and whose edges point *downstream*. Each node i carries:

- a_i [L^2], the area of the sub-basin uniquely defined by sample site i ;
- ϕ_i [$ML^{-2}T^{-1}$], the material export rate of that sub-basin;
- $q_i = a_i\phi_i > 0$ [MT^{-1}], the material flux the sub-basin generates;
- c_i , the source concentration of the tracer in material exported by sub-basin i — the unknown;
- d_i , the concentration observed at the sample site — the datum.

Write $U_i \subseteq N$ for the set of nodes upstream of i *including i itself*, and $A(i)$ for the set of immediate upstream neighbours of i (its predecessors in G). We make one structural assumption throughout, which the D8 flow model guarantees:

(A1) Every node has at most one downstream neighbour, so G is a forest of in-trees.

Assumption (A1) is what makes the sets $\{U_p\}_{p \in A(i)}$ pairwise disjoint, which we use in Proposition 5. It does not require the network to be connected: disjoint river basins may be solved together, exactly as in the parent study.

Two derived quantities recur. The **total flux** passing site i is

2 THE FORWARD MODEL AS A MATRIX

2.1 Assembling $\mathbf{M}$

Conservation of tracer at steady state gives the observation at site i as a flux-weighted mixture of every source upstream of it, attenuated by decay along the way:

$$d_i = \frac{\sum_{j \in U_i} \alpha_{j \rightarrow i} q_j c_j}{\sum_{j \in U_i} q_j} = \frac{1}{Q_i} \sum_{j \in U_i} \alpha_{j \rightarrow i} q_j c_j. \quad (3)$$

Note that decay enters the numerator only. Tracer is lost in transit, but the *carrier* — the water or sediment — is not, so the denominator is the undecayed total flux Q_i . Setting every $\alpha = 1$ recovers Equation 3 of Barnes and Lipp (2024).

Following the parent appendix, define the 0–1 matrix Θ by $\Theta_{ij} = 1$ if and only if $j \in U_i$; that is, Θ is the path matrix of G with a self-loop added at every node. Equation (3) is then linear in $\mathbf{c}$, with

$$M_{ij} = \frac{\Theta_{ij} \alpha_{j \rightarrow i} q_j}{\sum_{k=1}^n \Theta_{ik} q_k} = \frac{\Theta_{ij} \alpha_{j \rightarrow i} q_j}{Q_i}, \quad \mathbf{d} = \mathbf{M}\mathbf{c}. \quad (4)$$

For a conservative tracer this is exactly Equation 11 of the parent appendix.

2.2 Structure of $\mathbf{M}$

$\mathbf{M}$ is $n \times n$: the sample network defines exactly one sub-basin per sample site, so there are as many unknowns as data. This squareness is the pivot on which everything below turns, and it is worth stating plainly that it is a modelling choice, not a coincidence — it is what it means to attribute one source concentration to each sampled sub-basin.

Lemma 1 (Row sums). *Every row of $\mathbf{M}$ is non-negative with $\sum_j M_{ij} \leq 1$, with equality for all i if and only if the tracer is conservative.*

Proof. Non-negativity is immediate from (4) since $q_j > 0$, $\alpha > 0$, $\Theta \in \{0, 1\}$. For the sum,

$$\sum_{j=1}^n M_{ij} = \frac{1}{Q_i} \sum_{j \in U_i} \alpha_{j \rightarrow i} q_j \leq \frac{1}{Q_i} \sum_{j \in U_i} q_j = \frac{Q_i}{Q_i} = 1,$$

using $\alpha_{j \rightarrow i} \leq 1$, which holds because every $k_e L_e \geq 0$. Equality for a given i requires $\alpha_{j \rightarrow i} = 1$ for all $j \in U_i$, i.e. no decay anywhere upstream of i ; equality for all i requires it everywhere. $\square$

For a conservative tracer, then, $\mathbf{M}$ is **row-stochastic**

3 THE INVERSE IN CLOSED FORM

Proposition 3 says $\mathbf{c} = \mathbf{M}^{-1}\mathbf{d}$ exists and is unique. We now show it need never be computed by matrix inversion: it is a local differencing stencil on the network, computable in one sweep.

It is convenient to work with the **tracer flux** passing site i ,

$$T_i = Q_i d_i = \sum_{j \in U_i} \alpha_{j \rightarrow i} q_j c_j, \quad (7)$$

which is just (3) cleared of its denominator. T_i has units of tracer mass per time: it is the amount of tracer passing the sample site per unit time.

Proposition 5 (Closed-form inverse). *For every node i ,*

$$c_i = \frac{1}{q_i} \left(Q_i d_i - \sum_{p \in A(i)} \alpha_{p \rightarrow i} Q_p d_p \right). \quad (8)$$

Proof. Every node in U_i other than i lies upstream of exactly one immediate neighbour $p \in A(i)$, and by (A1) the sets U_p for distinct $p \in A(i)$ are pairwise disjoint. Hence

$$U_i = \{i\} \sqcup \bigsqcup_{p \in A(i)} U_p,$$

a disjoint union. Now fix $p \in A(i)$ and $j \in U_p$. The unique path $j \rightarrow i$ is the path $j \rightarrow p$ followed by the single edge $p \rightarrow i$, so the attenuation factorises:

$$\alpha_{j \rightarrow i} = \alpha_{j \rightarrow p} \alpha_{p \rightarrow i}. \quad (9)$$

Substituting the partition and (9) into (7), and using $\alpha_{i \rightarrow i} = 1$ for the $j = i$ term,

$$\begin{aligned} T_i &= q_i c_i + \sum_{p \in A(i)} \sum_{j \in U_p} \alpha_{j \rightarrow i} q_j c_j \\ &= q_i c_i + \sum_{p \in A(i)} \alpha_{p \rightarrow i} \underbrace{\sum_{j \in U_p} \alpha_{j \rightarrow p} q_j c_j}_{= T_p} \\ &= q_i c_i + \sum_{p \in A(i)} \alpha_{p \rightarrow i} T_p. \end{aligned} \quad (10)$$

Equation (10) is a statement of tracer conservation at node i : the tracer leaving equals the tracer generated locally plus the (attenuated) tracer delivered by the immediate tributaries. Solving for c

Corollary 7 (Worked example). *For the six-node network of Barnes and Lipp (2024, Figure 2), with $A(4) = \{1, 2\}$, $A(5) = \{3\}$, $A(6) = \{4, 5\}$ and no decay, Equation (8) gives*

$$\begin{aligned} c_1 &= d_1, & c_2 &= d_2, & c_3 &= d_3, \\ c_4 &= \frac{Q_4 d_4 - Q_1 d_1 - Q_2 d_2}{q_4}, \\ c_5 &= \frac{Q_5 d_5 - Q_3 d_3}{q_5}, \\ c_6 &= \frac{Q_6 d_6 - Q_4 d_4 - Q_5 d_5}{q_6}, \end{aligned}$$

with $Q_1 = q_1$, $Q_2 = q_2$, $Q_3 = q_3$, $Q_4 = q_1 + q_2 + q_4$, $Q_5 = q_3 + q_5$ and $Q_6 = \sum_{i=1}^6 q_i$. Headwater sites are recovered exactly and trivially; interior sites are recovered by differencing.

3.1 Noise amplification

Differencing is the source of all the difficulty. Suppose the observations carry independent errors of standard deviation σ_i . Propagating them through (8) (anticipating Section 8, and using independence),

$$\text{Var}(c_i) = \frac{Q_i^2 \sigma_i^2 + \sum_{p \in A(i)} \alpha_{p \rightarrow i}^2 Q_p^2 \sigma_p^2}{q_i^2}. \quad (12)$$

Every term is scaled by q_i^{-2} . Define the **amplification factor**

$$\kappa_i = \frac{Q_i}{q_i} = \frac{1}{M_{ii}} \geq 1, \quad (13)$$

the ratio of the total flux passing site i to the flux its own sub-basin generates. If site i sits immediately downstream of another site, with little incremental drainage area between them, then $q_i \ll Q_i$, κ_i is large, and (8) subtracts two nearly equal large numbers to obtain a small one. This is catastrophic cancellation, and it is an intrinsic property of the sampling design, not of the algorithm: the data simply do not constrain that sub-basin well. In practice κ_i , being free to compute, is the single most useful diagnostic of whether a linear inversion is worth attempting on a given network.

Corollary 8 (When positivity binds). *Since concentrations must be non-negative, the unregularized solution is physically admissible at node i if and only if*

$$Q_i d_i \geq \sum_{p \in A(i)} \alpha_{p \rightarrow i}$$

1. **Weighting is inert when $\lambda = 0$.** No choice of $\mathbf{W}$ — including $\mathbf{W} = \mathbf{C}_d^{-1}$, the statistically correct one — changes the answer. There is nothing to trade off when the model can fit every datum exactly. Weighting only becomes consequential once a competing term is added to the objective (Section 7).
2. **The problem interpolates the noise.** A zero residual means the estimate reproduces the observations exactly, errors included. Combined with the amplification of Section 3.1, this is why the raw inversion is unusable on realistic data.
3. **The constraint is the only thing that can bite.** By Corollary 8, (15) departs from $\mathbf{M}^{-1}\mathbf{d}$ precisely when (14) fails somewhere.

Remark 10 (Bounds). The parent appendix writes the constraint as $0 < c_i < 1$, appropriate for a tracer measured as a mass fraction. Two remarks. First, a strict inequality cannot be imposed in a convex program — the feasible set would not be closed and the infimum need not be attained — so one uses $c_i \geq 0$. Second, the upper bound is unit-dependent (1 for a fraction, 10^6 for mg kg^{-1}) whereas the lower bound is not. Since $\mathbf{M}$ is row-substochastic, an estimate exceeding the physical ceiling indicates that the model has already failed, and is more useful surfaced as a diagnostic than silently clipped. Below we impose $c_i \geq 0$ only.

5 SCALE INVARIANCE AND NORMALISATION

Proposition 11 (Homogeneity). *The unregularized solution map $\mathbf{d} \mapsto \mathbf{c}$ is homogeneous of degree one: $\mathbf{c}(t\mathbf{d}) = t\mathbf{c}(\mathbf{d})$ for any $t > 0$. Moreover $\mathbf{M}$ is invariant under a common rescaling of the fluxes, $q_j \mapsto sq_j$.*

Proof. The first claim is immediate from linearity, $\mathbf{c} = \mathbf{M}^{-1}\mathbf{d}$. For the second, replacing q_j by sq_j in (4) multiplies numerator and denominator by s . $\square$

Homogeneity licenses the mean normalisation used in the implementation. Writing $\bar{d} = n^{-1} \sum_i d_i$ and solving with $\tilde{\mathbf{d}} = \mathbf{d}/\bar{d}$, the recovered sources are $\mathbf{c} = \bar{d} \tilde{\mathbf{c}}$ exactly. This changes nothing mathematically; its purpose is numerical, keeping all quantities near unity so that solver tolerances — which are partly absolute — behave uniformly across tracers whose concentrations may differ by many orders of magnitude. It has the further benefit of rendering the regularization strength introduced below dimensionless, and hence transferable between tracers and datasets.

An identical argument applies to the sub-basin areas, which the implementation divides by their mean, and to covarian

7 REGULARIZATION

Section 4 showed that the unregularized problem interpolates the noise. The remedy is Tikhonov regularization (Hansen 1998; Hoerl and Kennard 1970; Tikhonov 1963): add a penalty expressing a preference among models.

We penalise the **variance of the model about its own mean**. Let

$$\mathbf{P} = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top \quad (20)$$

be the centering projector, so that $(\mathbf{P}\mathbf{c})_i = c_i - \bar{c}$. The full problem is

$$\begin{array}{ll} \underset{\mathbf{c} \in \mathbb{R}^n}{\text{minimize}} & J(\mathbf{c}) = (\mathbf{M}\mathbf{c} - \mathbf{d})^\top \mathbf{C}_d^{-1}(\mathbf{M}\mathbf{c} - \mathbf{d}) + \lambda \|\mathbf{P}\mathbf{c}\|_2^2 \\ \text{subject to} & \mathbf{c} \geq 0. \end{array} \quad (21)$$

Note that this penalises departure from the mean of the *model*, not, as in the parent study, from the mean of the *observations*. That substitution was forced there by the requirement that the objective be expressible in disciplined convex form (Boyd and Vandenberghe 2004; Diamond and Boyd 2016); the linear formulation is under no such constraint and can use the more natural quantity.

Lemma 13 (Properties of $\mathbf{P}$). *$\mathbf{P}$ is symmetric and idempotent, hence an orthogonal projector; it is positive semi-definite with eigenvalue 0 on $\text{span}\{\mathbf{1}\}$ and eigenvalue 1 on its orthogonal complement; and*

$$\|\mathbf{P}\mathbf{c}\|_2^2 = \mathbf{c}^\top \mathbf{P}^\top \mathbf{P} \mathbf{c} = \mathbf{c}^\top \mathbf{P} \mathbf{c} = \sum_{i=1}^n (c_i - \bar{c})^2. \quad (22)$$

Proof. Symmetry is clear. For idempotency, using $\mathbf{1}^\top \mathbf{1} = n$,

$$\mathbf{P}^2 = \mathbf{I} - \frac{2}{n}\mathbf{1}\mathbf{1}^\top + \frac{1}{n^2}\mathbf{1}(\mathbf{1}^\top \mathbf{1})\mathbf{1}^\top = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top = \mathbf{P}.$$

Then $\mathbf{P}^\top \mathbf{P} = \mathbf{$

Two features of (26) deserve emphasis. First, it is exactly linear in $\mathbf{d}$, with **no constant offset**. This is a consequence of penalising $\|\mathbf{P}\mathbf{c}\|^2$, which is homogeneous in $\mathbf{c}$; a penalty $\|\mathbf{c} - \mathbf{c}_0\|^2$ towards a fixed reference $\mathbf{c}_0 \neq \mathbf{0}$ would contribute an affine term and complicate everything downstream. Second, Proposition 16 gives a practical recipe: solve the problem, and if the answer is non-negative, report the analytical $\mathbf{R}_\lambda \mathbf{d}$. The estimate and its covariance then provably come from the same linear operator, which is what makes the error propagation of Section 8 meaningful. When the constraint is active the estimator is only piecewise linear, and that guarantee is lost.

7.1 Limits

Proposition 17 (Small λ). $\mathbf{R}_0 = \mathbf{M}^{-1}$, recovering Section 3.

Proof. Set $\lambda = 0$ in (26): $\mathbf{R}_0 = (\mathbf{M}^\top \mathbf{C}_d^{-1} \mathbf{M})^{-1} \mathbf{M}^\top \mathbf{C}_d^{-1} = \mathbf{M}^{-1}$, as in Proposition 9. $\square$

Proposition 18 (Large λ). As $\lambda \rightarrow \infty$, $\hat{\mathbf{c}} \rightarrow t^* \mathbf{1}$ with

$$t^* = \frac{\mathbf{m}^\top \mathbf{C}_d^{-1} \mathbf{d}}{\mathbf{m}^\top \mathbf{C}_d^{-1} \mathbf{m}}, \quad \mathbf{m} = \mathbf{M} \mathbf{1}. \quad (27)$$

If in addition the tracer is conservative and $\mathbf{C}_d = \sigma^2 \mathbf{I}$, then $t^* = \bar{d}$, the arithmetic mean of the observations.

Proof. The penalty term $\lambda \|\mathbf{P}\mathbf{c}\|^2$ diverges unless $\mathbf{P}\mathbf{c} \rightarrow \mathbf{0}$, i.e. unless $\mathbf{c}$ approaches $\mathcal{N}(\mathbf{P}) = \text{span}\{\mathbf{1}\}$. Writing $\mathbf{c} = t\mathbf{1}$ the penalty vanishes identically and the problem reduces to minimising $(t\mathbf{m} - \mathbf{d})^\top \mathbf{C}_d^{-1} (t\mathbf{m} - \mathbf{d})$ over the scalar t . Differentiating and setting to zero gives t^* as stated. For a conservative tracer Lemma 1 gives $\mathbf{m} = \mathbf{M} \mathbf{1} = \mathbf{1}$, whence with $\mathbf{C}_d = \sigma^2 \mathbf{I}$ we get $t^* = \mathbf{1}^\top \mathbf{d} / \mathbf{1}^\top \mathbf{1} = \bar{d}$. $\square$

So the regularization path runs from the exact,

No estimator, weighted or otherwise, can do better than $\mathbf{M}^{-1}\mathbf{d}$ in the unbiased class. Section 4 already told us weighting cannot help; this tells us why nothing else can either. The only way to improve on (12) is to accept bias — which is exactly what regularization does.

8.2 Bias, resolution, and the two covariances

Proposition 22 (Bias and resolution). $\mathbb{E}[\hat{\mathbf{c}}] = \mathbf{R}_\lambda \mathbf{M} \mathbf{c}_{\text{true}}$, so the estimator has bias

$$\mathbf{b} = (\mathbf{R}_\lambda \mathbf{M} - \mathbf{I}) \mathbf{c}_{\text{true}} = -\lambda \mathbf{A}_\lambda^{-1} \mathbf{P} \mathbf{c}_{\text{true}}, \quad (31)$$

and resolution matrix

$$\mathbf{R}_\lambda \mathbf{M} = \mathbf{I} - \lambda \mathbf{A}_\lambda^{-1} \mathbf{P}. \quad (32)$$

The mean-squared error is $\mathbb{E}[(\hat{\mathbf{c}} - \mathbf{c}_{\text{true}})(\cdot)^T] = \mathbf{C}_c + \mathbf{b}\mathbf{b}^T$.

Proof. Substituting $\mathbf{d} = \mathbf{M} \mathbf{c}_{\text{true}} + \boldsymbol{\varepsilon}$ with $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ into (26) gives the first claim. Then

$$\mathbf{R}_\lambda \mathbf{M} - \mathbf{I} = \mathbf{A}_\lambda^{-1} \mathbf{M}^T \mathbf{C}_d^{-1} \mathbf{M} - \mathbf{I} = \mathbf{A}_\lambda^{-1} (\mathbf{M}^T \mathbf{C}_d^{-1} \mathbf{M} - \mathbf{A}_\lambda) = -\lambda \mathbf{A}_\lambda^{-1} \mathbf{P},$$

using (25). The MSE decomposition is the standard bias–variance identity (Aster et al. 2018; Hoerl and Kennard 1970). $\square$

The bias vanishes when $\lambda = 0$, and also whenever $\mathbf{P} \mathbf{c}_{\text{true}} = \mathbf{0}$, i.e. when the true source field really is uniform — the penalty is then telling the truth and costs nothing. Since $\mathbf{c}_{\text{true}}$ is unknown, $\mathbf{b}$ cannot be evaluated; the resolution matrix (32) is the standard computable proxy (Menke 2012), equal to $\mathbf{I}$ at $\lambda = 0$ and degrading as λ grows. Its trace, $\text{tr}(\mathbf{R}_\lambda \mathbf$

Quantity	Expression
Mixing matrix	$M_{ij} = \Theta_{ij} \alpha_{j \rightarrow i} q_j / Q_i$
Exact inverse	$c_i = (Q_i d_i - \sum_{p \in A(i)} \alpha_{p \rightarrow i} Q_p d_p) / q_i$
Amplification	$\kappa_i = Q_i / q_i = 1 / M_{ii}$
Penalty operator	$\mathbf{P} = \mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top$
Normal matrix	$\mathbf{A}_\lambda = \mathbf{M}^\top \mathbf{C}_d^{-1} \mathbf{M} + \lambda \mathbf{P}$
Estimator	$\mathbf{R}_\lambda = \mathbf{A}_\lambda^{-1} \mathbf{M}^\top \mathbf{C}_d^{-1}$
Source covariance	$\mathbf{C}_c = \mathbf{R}_\lambda \mathbf{C}_d \mathbf{R}_\lambda^\top$
Prediction covariance	$\mathbf{C}_d = \mathbf{M} \mathbf{C}_c \mathbf{M}^\top$
Posterior covariance	$\mathbf{A}_\lambda^{-1}$
Resolution	$\mathbf{R}_\lambda \mathbf{M} = \mathbf{I} - \lambda \mathbf{A}_\lambda^{-1} \mathbf{P}$
Bias	$\mathbf{b} = -\lambda \mathbf{A}_\lambda^{-1} \mathbf{P} \mathbf{c}_{\text{true}}$

Table 1: **Principal results.** At $\lambda = 0$: $\mathbf{R}_0 = \mathbf{M}^{-1}$, the resolution matrix is $\mathbf{I}$, the bias vanishes, the two covariances coincide, and $\mathbf{C}_c$ attains the Cramér–Rao bound.

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